Task 2 Report — Multi-Object Tracking on MOT17

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Course/Competition: CV Competition (Task 2 – Multi-Object Tracking)

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1. Introduction

This report describes a baseline multi-object tracking (MOT) system evaluated on the MOT17 dataset. The objective is to track pedestrians across frames while maintaining consistent identities over time. Our approach uses a YOLOv8 detector to produce per-frame person detections and a lightweight tracker that associates detections across frames using Intersection-over-Union (IoU).

2. Methodology

2.1 Detection

We use YOLOv8 (pretrained weights yolov8s.pt) to detect the 'person' class in each frame. Detections below a confidence threshold are filtered out.

Key detection settings:

- Model: yolov8s.pt (Ultralytics)
- Class filter: person only
- Confidence threshold: 0.30

2.2 Tracking / Data Association

We implement a ByteTrack-style tracker that maintains active tracks and associates new detections to existing tracks using IoU. Matching is performed greedily by selecting the highest IoU pairs above a threshold. Unmatched detections start new tracks, and tracks that are not updated for a fixed number of frames are removed.

Key tracking settings:

- IoU threshold: 0.30
- Max age (frames without update before deletion): 30
- Matching: Greedy IoU matching

2.3 Output Format

Tracking outputs are saved in MOTChallenge format per sequence as a CSV-like text file with columns:
frame, id, x, y, w, h, score, -1, -1, -1

3. Experiments & Results

We evaluate on the MOT17 training sequences (DPM variants) and compute standard metrics using motmetrics: MOTA, MOTP, IDF1, and ID switches.

Sequence	MOTA	MOTP	IDF1	ID switches	# frames
MOT17-02-DPM	0.106756	0.238448	0.246455	117	600
MOT17-04-DPM	0.179279	0.193692	0.286158	108	1050
MOT17-05-DPM	-1.006240	0.257644	0.304642	266	837
MOT17-09-DPM	0.045817	0.226474	0.318317	110	525
MOT17-10-DPM	-0.171060	0.256542	0.244694	379	654
MOT17-11-DPM	-0.235471	0.170854	0.394881	50	900
MOT17-13-DPM	-0.466488	0.262849	0.140722	709	750

Overall (frame-weighted):

Frames: 5316
MOTA: -0.233171
MOTP: 0.225673
IDF1: 0.280550
ID switches: 1739

4. Discussion

The baseline tracker is lightweight and fast because it uses only IoU for association. However, it is sensitive to detection quality and struggles during occlusions and crowded scenes, which can increase false negatives/positives and cause frequent ID switches. Negative MOTA indicates that the combined errors (false positives, false negatives, ID switches) exceed correct matches.

5. Qualitative Results (Videos / Screenshots)

Generated overlay videos (bounding boxes + track IDs) were saved under:
task2_tracking\results\videos\

- MOT17-02-DPM.mp4
- MOT17-04-DPM.mp4

The screenshots below show example frames from the generated tracking visualization (MOT17-02-DPM), demonstrating ID persistence across frames and typical failure cases (occlusions / crowded regions).

Figure 1: MOT17-02-DPM – example tracking frame (IDs + boxes overlaid).

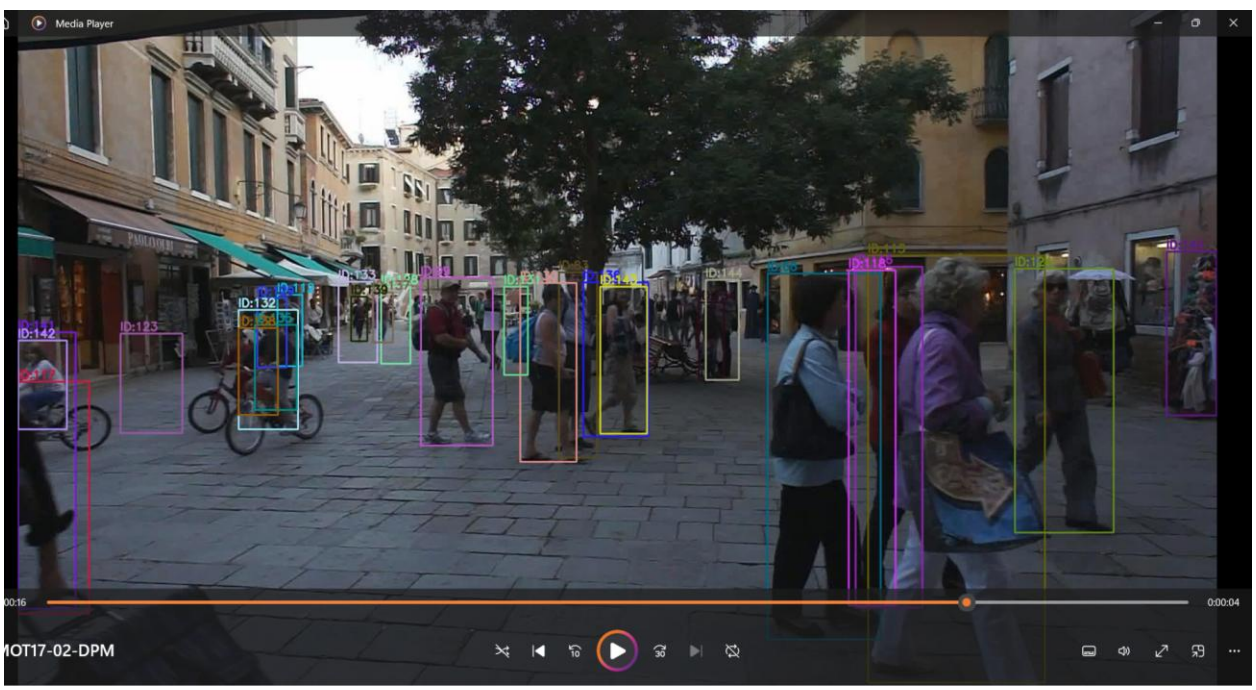
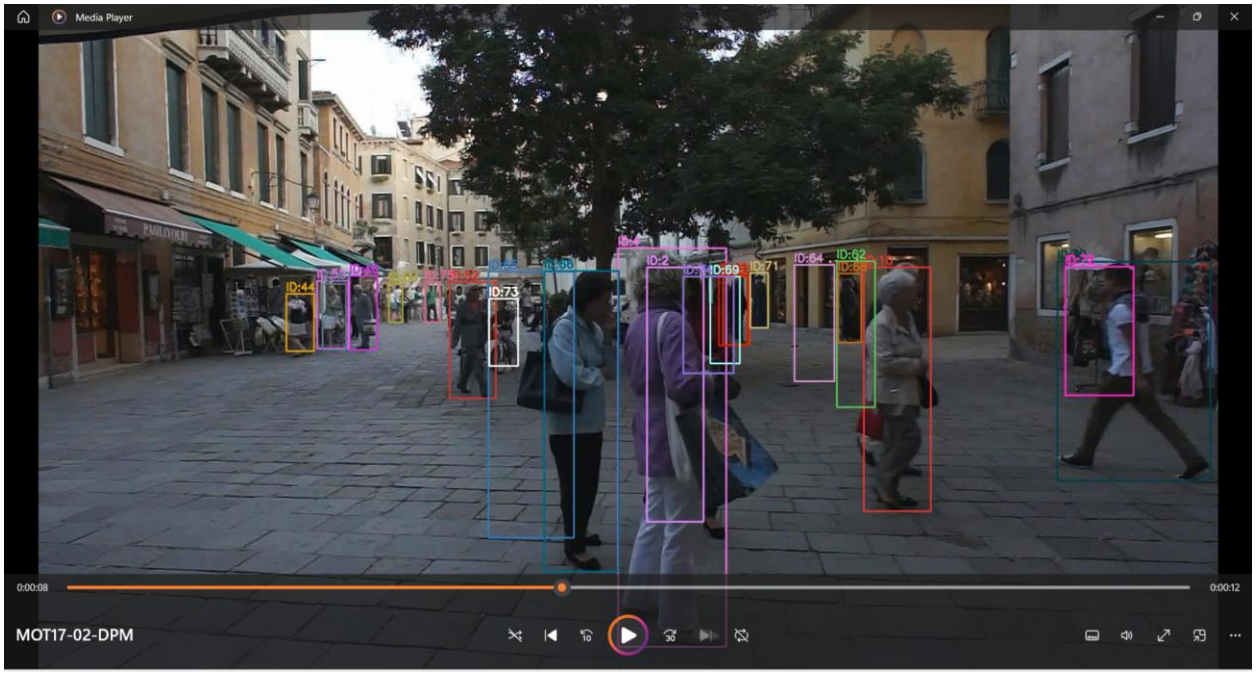


Figure 2: MOT17-02-DPM – example tracking frame during crowd/occlusion.



1) Activate environment
  yolo_env\Scripts\activate

2) Run tracking + evaluation (generates per-sequence tracks + metrics CSV)
  python task2_tracking\code\task2_complete.py

3) (Optional) Compute overall (frame-weighted) summary from the CSV
  python task2_tracking\code\make_overall.py

4) Generate visualization video(s)
  python task2_tracking\code\make_video_from_tracks.py --seq datasets\MOT17\train\MOT17-02-DPM --tracks task2_tracking\results\tracks\MOT17-02-DPM.txt --out task2_tracking\results\videos\MOT17-02-DPM.mp4
  python task2_tracking\code\make_video_from_tracks.py --seq datasets\MOT17\train\MOT17-04-DPM --tracks task2_tracking\results\tracks\MOT17-04-DPM.txt --out task2_tracking\results\videos\MOT17-04-DPM.mp4

5) Outputs produced
  task2_tracking\results\tracks*.txt
  task2_tracking\results\metrics\tracking_metrics.csv
  task2_tracking\results\videos*.mp4

6) Package submission
  powershell Compress-Archive -Path task2_tracking -DestinationPath task2_tracking_submission.zip -Force

6. Conclusion & Future Work

This project demonstrates a working end-to-end MOT pipeline with detection, tracking, and evaluation. Future improvements include:

- Using a stronger or fine-tuned detector (trained on MOT)
- Adding motion prediction (Kalman filter)
- Adding appearance features (e.g., DeepSORT/BoT-SORT) to reduce ID switches
- Hyperparameter tuning (confidence/IoU thresholds, max_age)

Appendix: How to Run

1) Activate environment
2) Ensure MOT17 is under datasets/MOT17/train/<sequence>/img1
3) Run:
   python task2_tracking\code\task2_complete.py
4) Outputs:
   task2_tracking\results\tracks*.txt
   task2_tracking\results\metrics\tracking_metrics.csv